

Strategy that know whether one non-goal state is "more promising" than another are called informed search or heuristic search strategies

Limitations of uninformed search

- ↳ 8-puzzle,

- ↳ Avg solution cost about 22 steps

- ↳ branching factor ~ 3

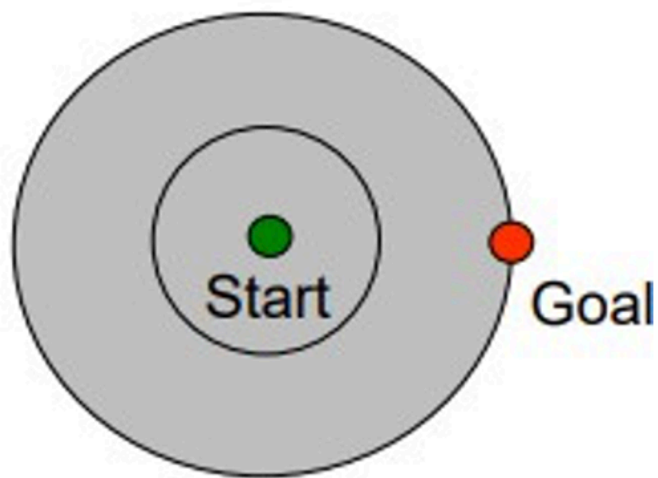
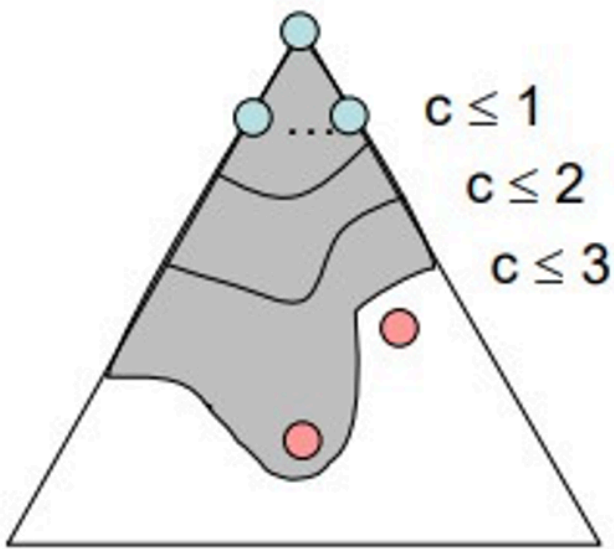
Example,

For $d=12$, IDS expands 3.6 million stages on average.

UCS issues

- ↳ Explores options in every direction

- ↳ No information about goal direction

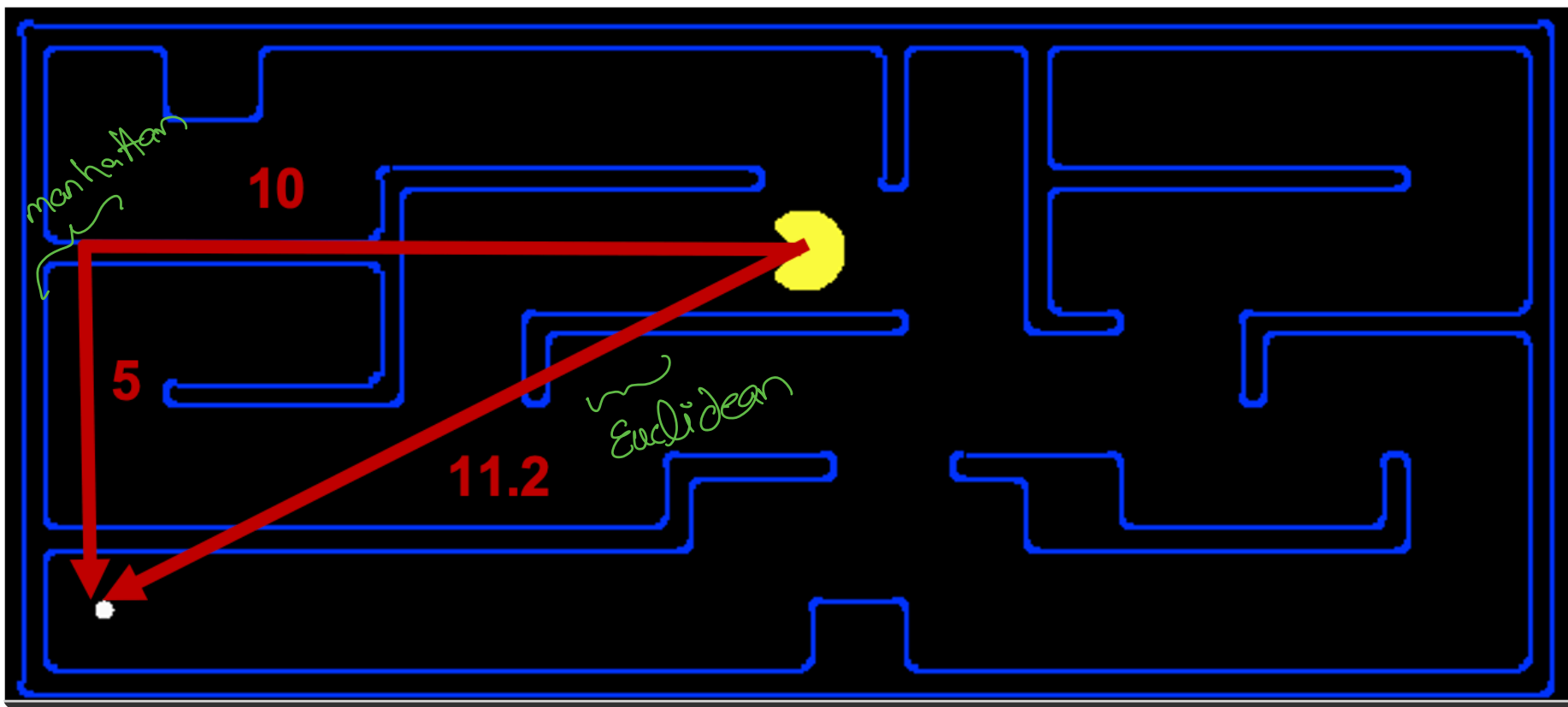


Search heuristics:

- ↳ A function that estimates how close a goal state is.

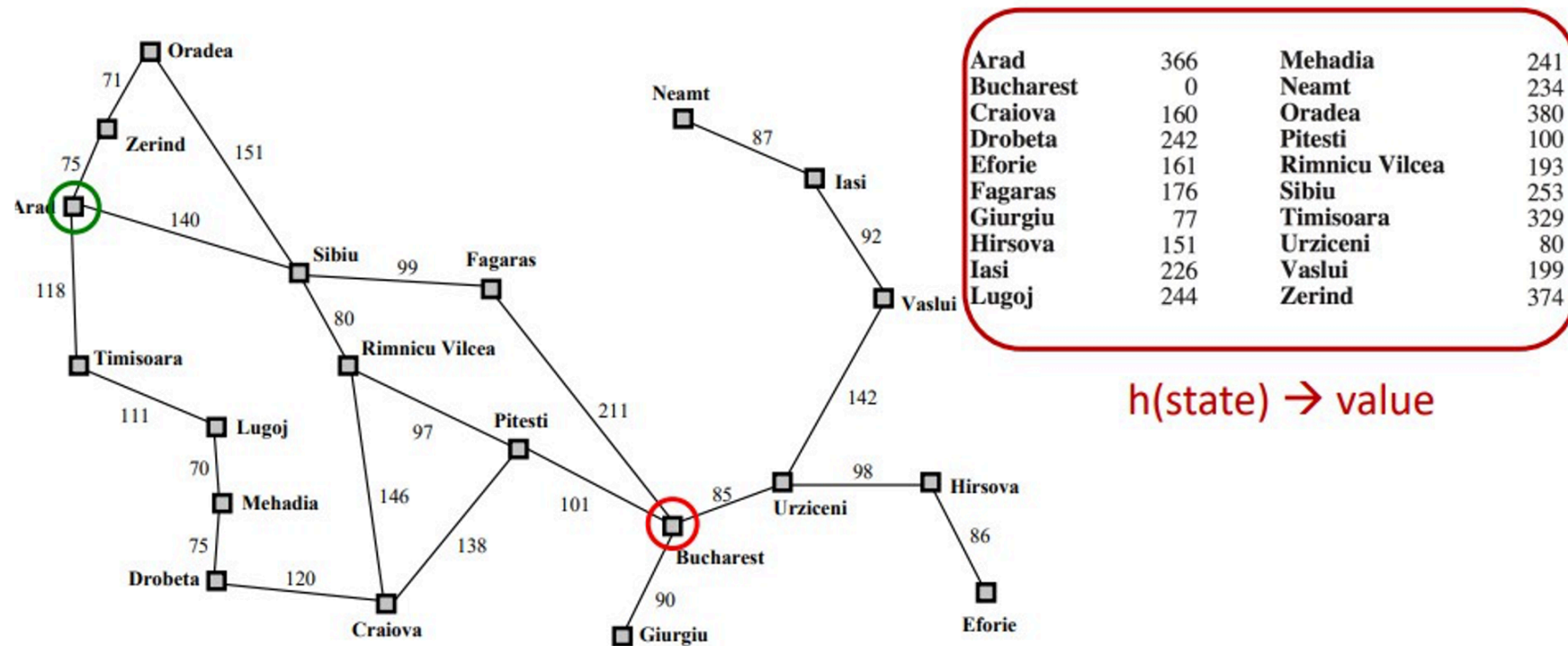
- ↳ Designed for a particular search problem

$h(n)$ = estimated cost of the cheapest path from node n to goal state.



Example,

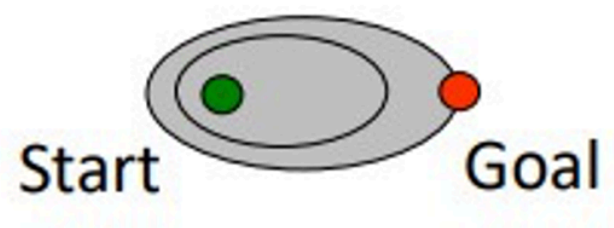
Example: Euclidean distance to Bucharest



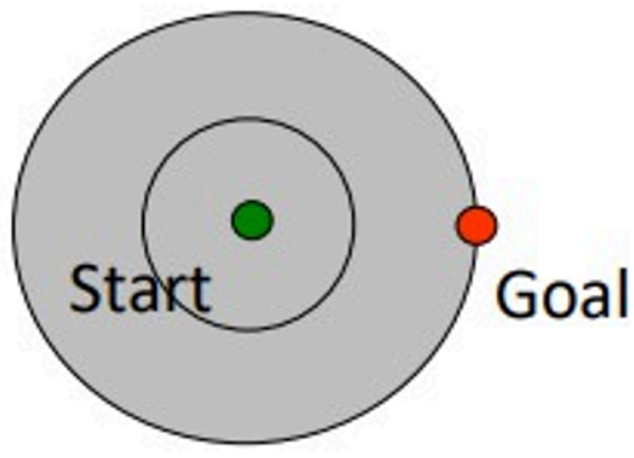
$h(\text{state}) \rightarrow \text{value}$

Effect of heuristics

Guide search towards the goal instead of all over the place.

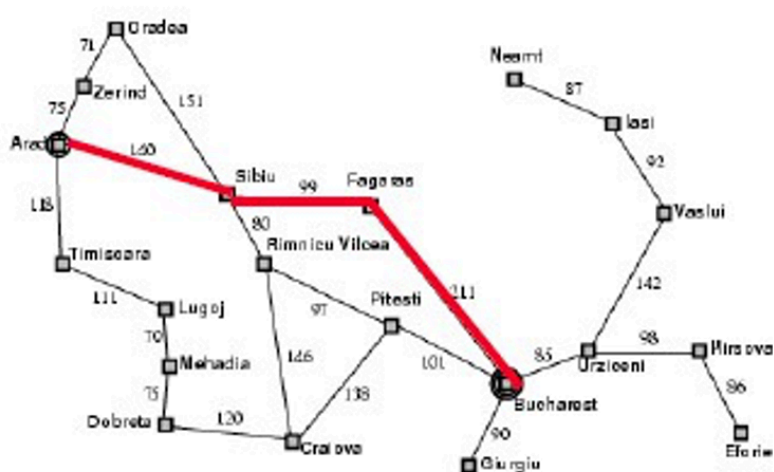
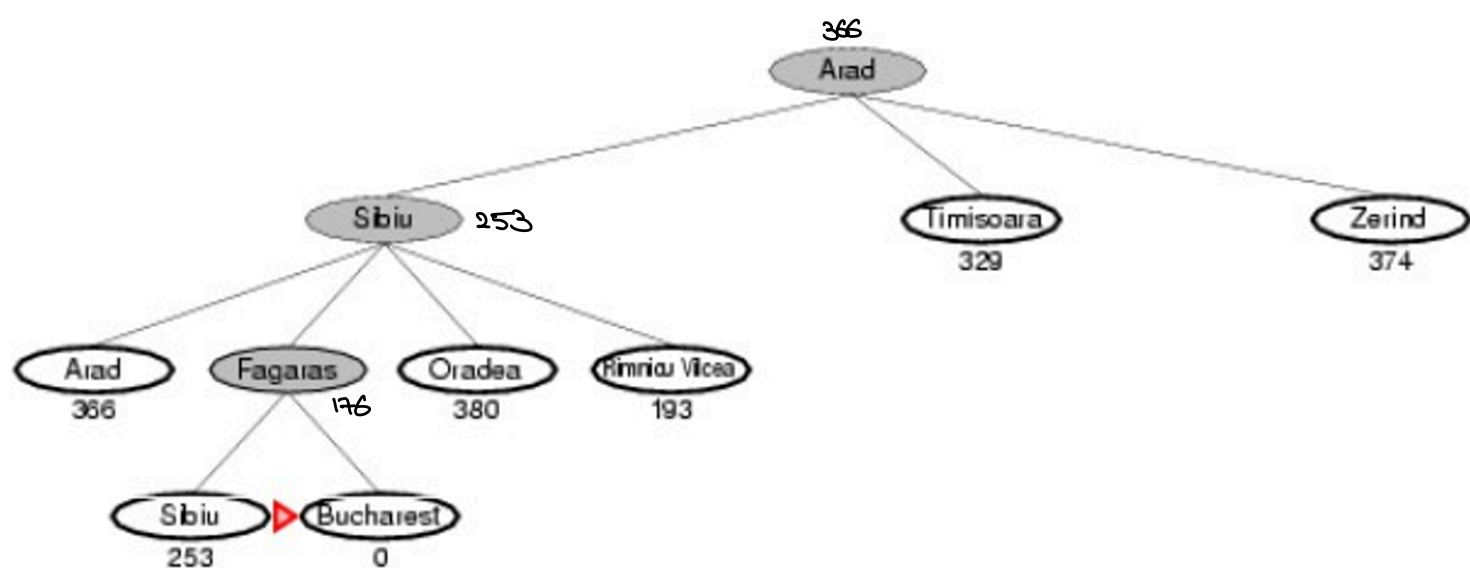


Informed

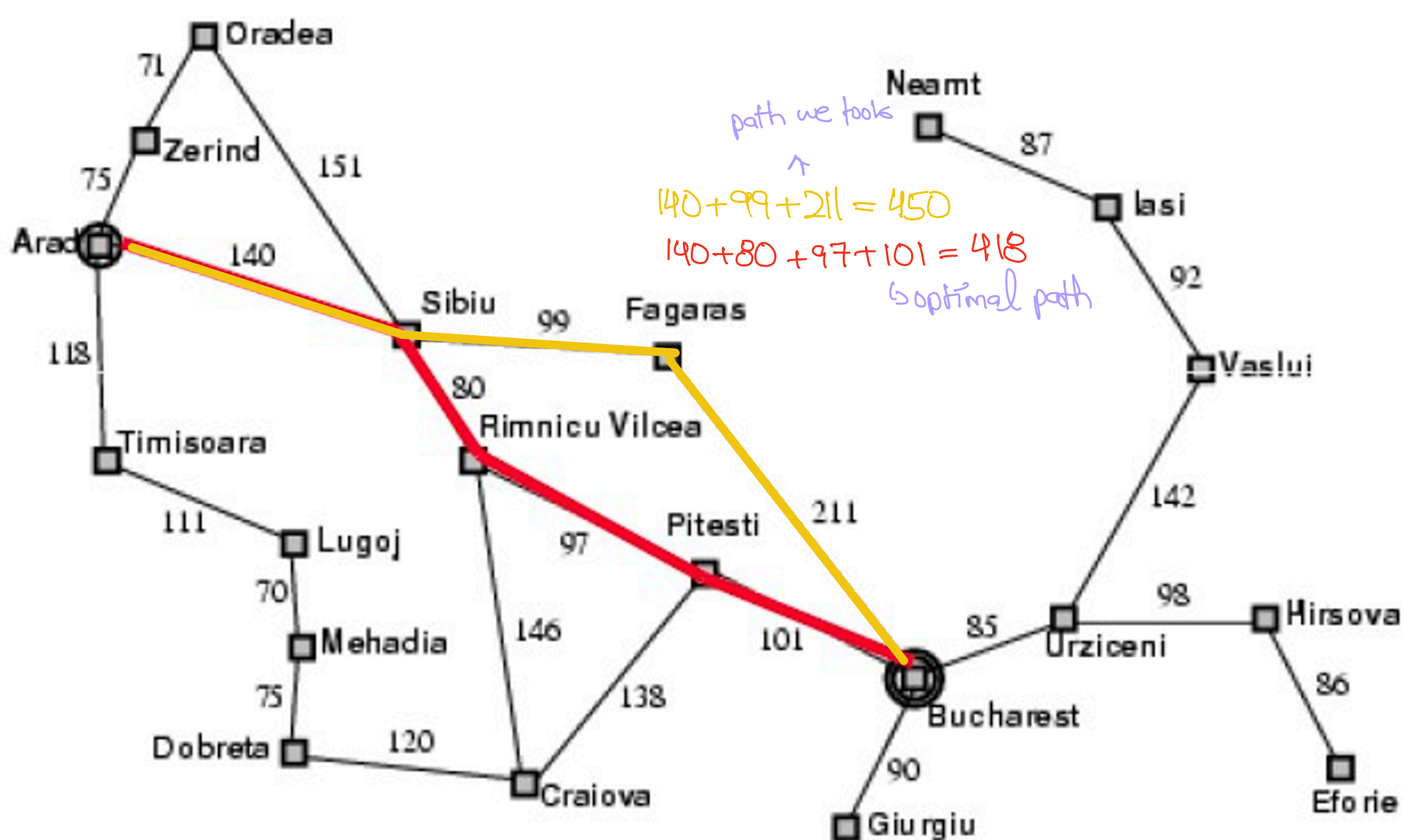


Uninformed

Greedy search



but optimal path is



Properties:

Time complexity: $O(b^m)$

Space complexity: $O(b^m)$

Completeness: No, can get stuck in loops if visited nodes are not tracked

Optimal: No

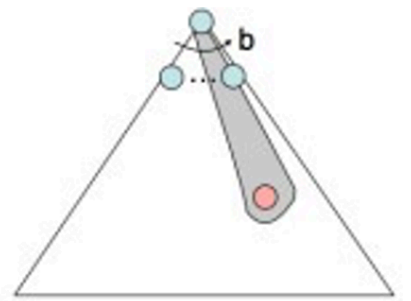
Strategy: expand a node that you think is closest to a goal state

Heuristic: estimate of distance to nearest goal for each state

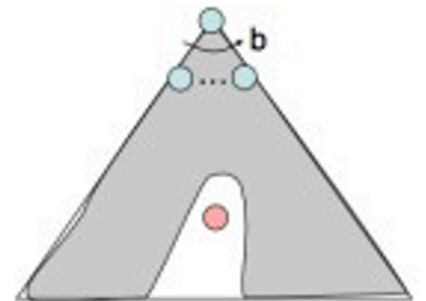
A common case:

Best-first takes you straight to the (wrong) goal as seen in example.

Worst-case: like a badly-guided DFS



(a) Greedy search on a good day :)



(b) Greedy search on a bad day :(

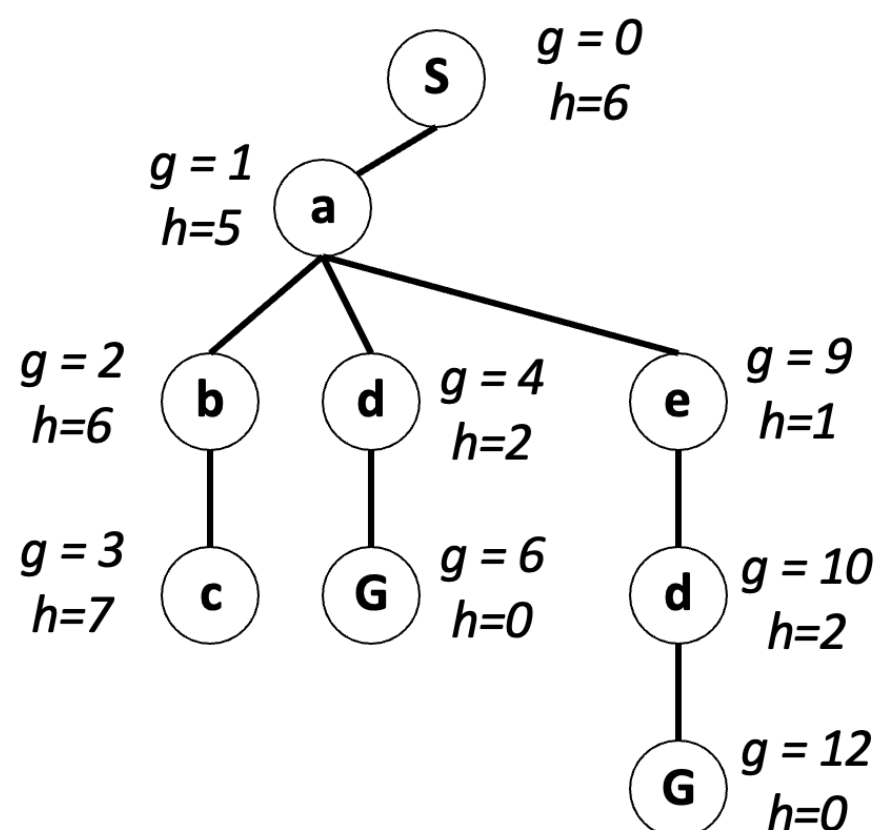
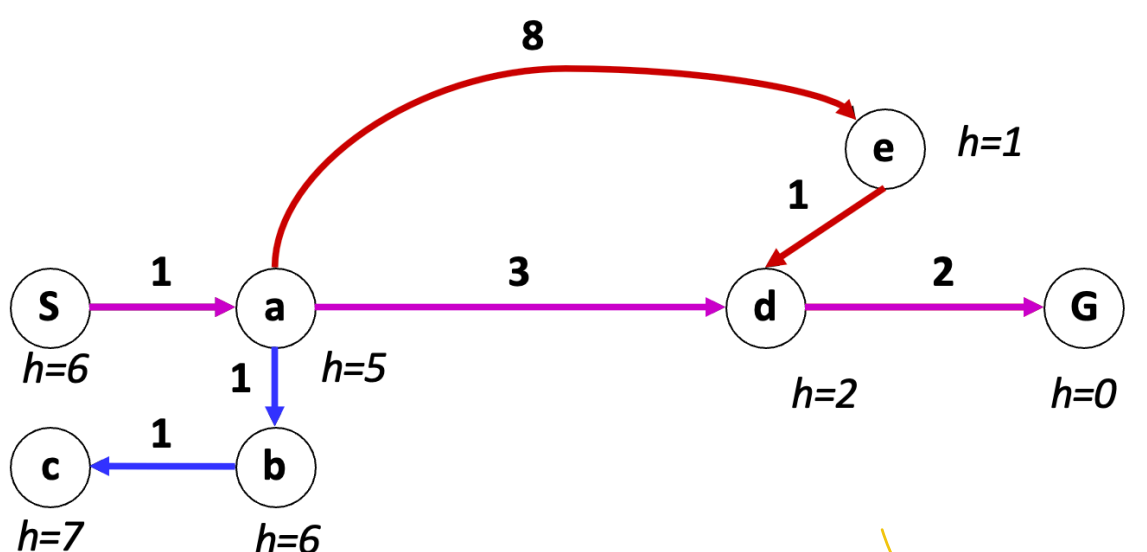
A* Search

Combining Greedy & UCS

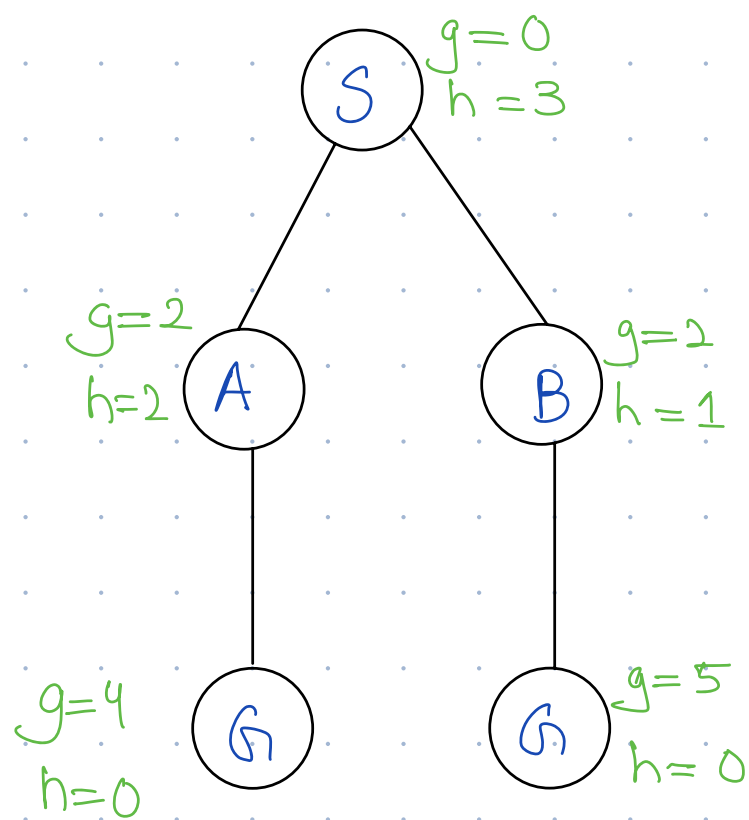
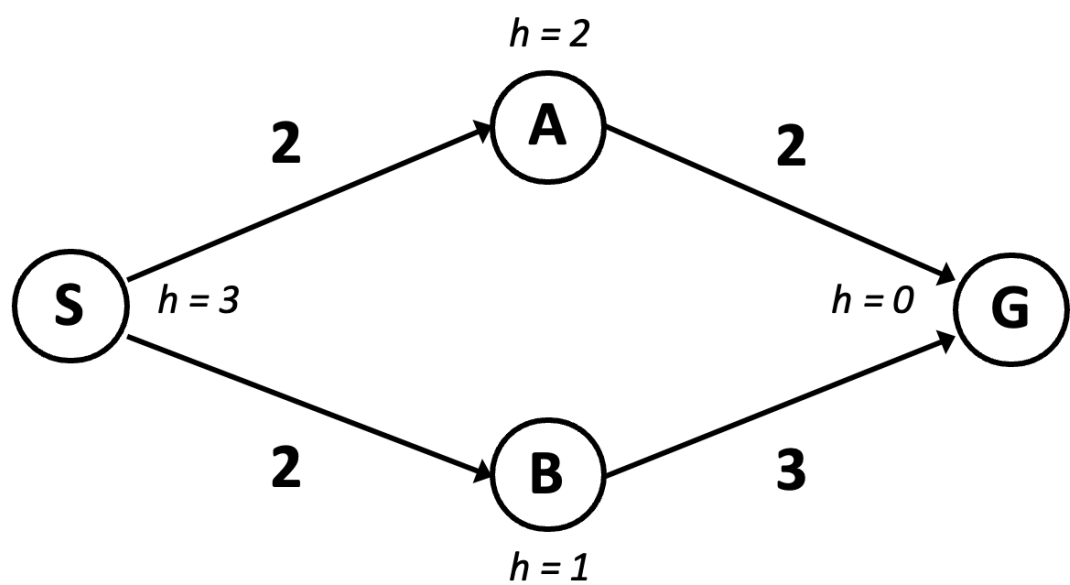
UCS orders by path cost/backward cost $g(n)$

Greedy orders by goal proximity/forward cost $h(n)$

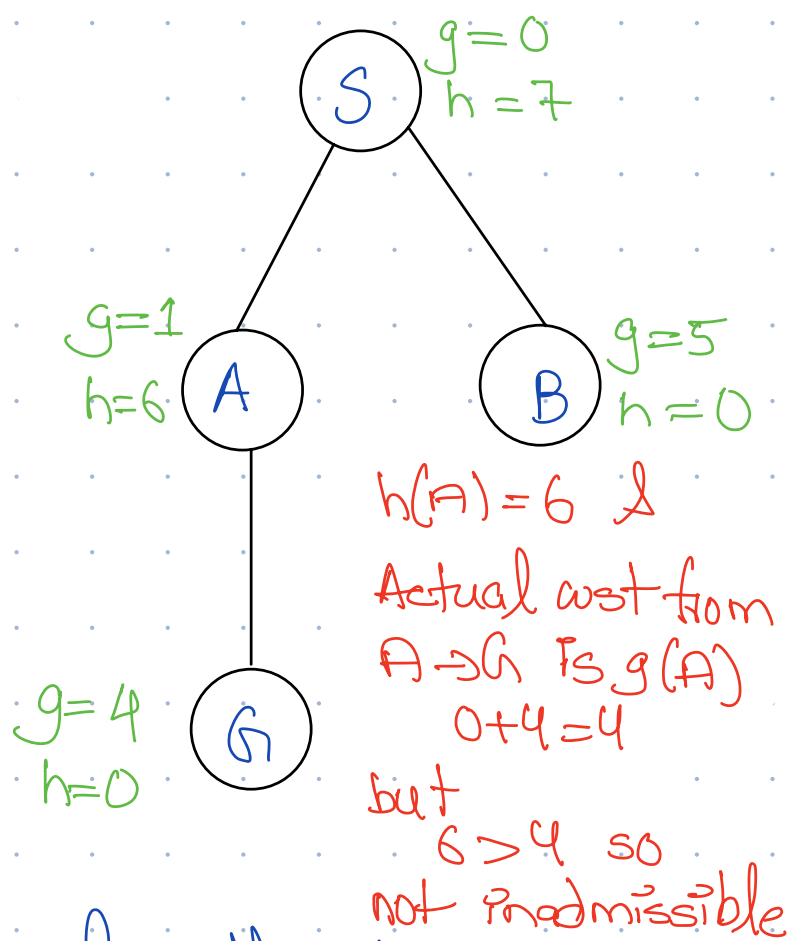
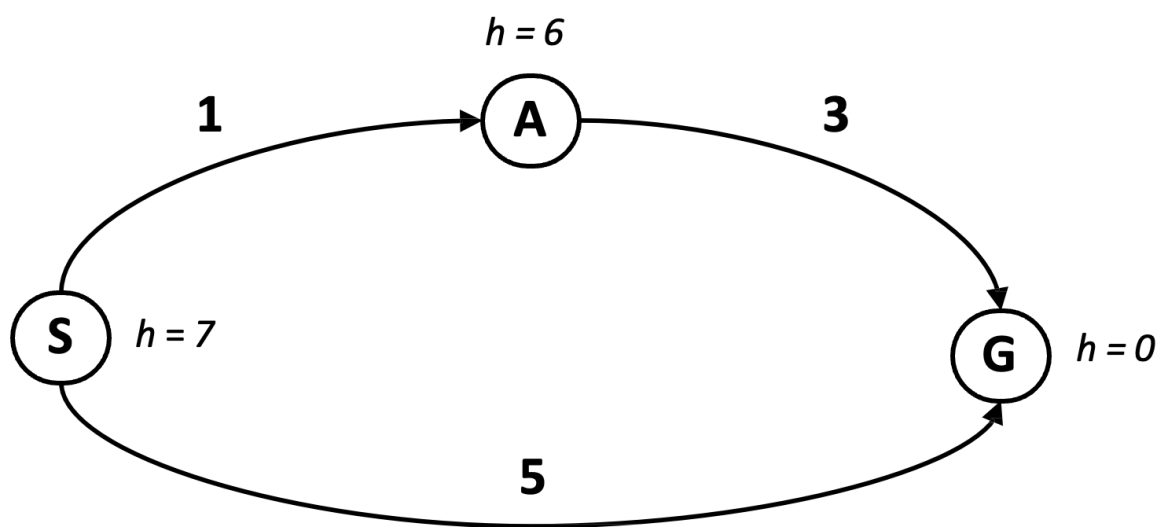
A* orders by $f(n) = g(n) + h(n)$



In case of a cycle, stop A* when we de-queue a goal state



Is A* optimal?



In both cases, we are choosing non-optimal paths because the actual goal cost $g(n)$ < estimated goal cost $h(n)$

Admissible Heuristics

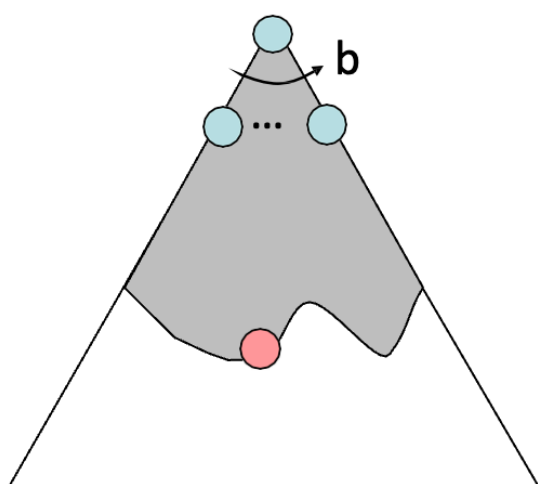
If $h(n) \leq h^*(n)$ i.e. estimated heuristic $h(n)$ is always less than equal to true cheapest cost from n to the goal.

$$\forall n \quad 0 \leq h(n) \leq h^*(n)$$

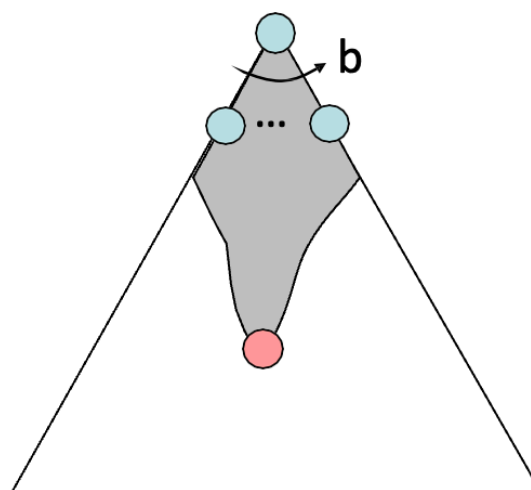
$h^*(n)$ is the true cost to nearest goal.

Thus, A* is admissible if h is admissible.

Uniform-Cost



A*



How to construct an admissible heuristic?

↳ Identify a relaxed version of the problem

↳ Where one or more constraints have been dropped.

↳ Problem with fewer restrictions for action

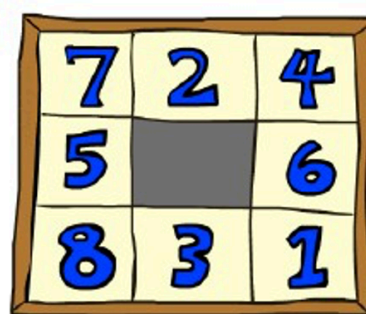
Robot: The agent can move through walls

8-puzzle: Tiles can move anywhere or can move to any adjacent square.

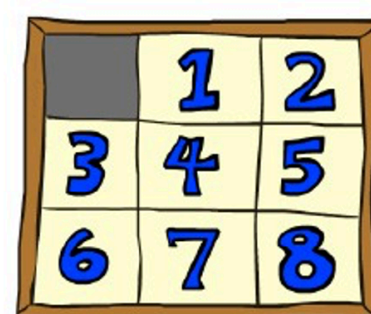
Example,

8 Puzzle I

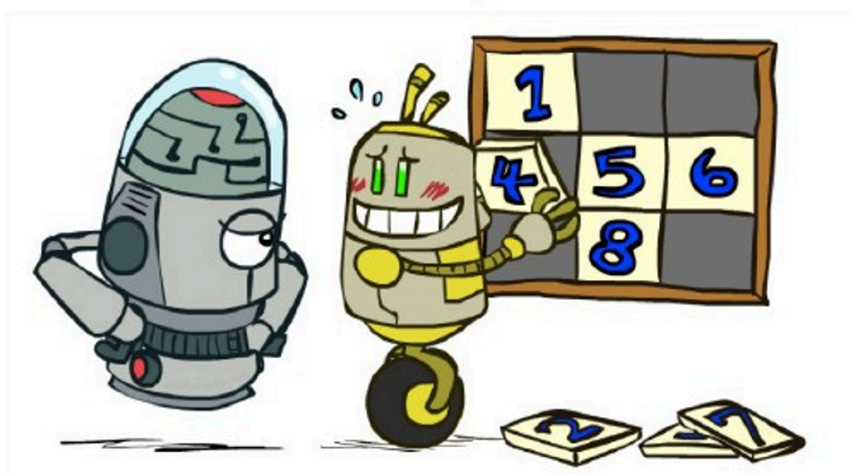
- Heuristic: Number of tiles misplaced
- Why is it admissible?
- $h(\text{start}) = 8$
- This is a *relaxed-problem* heuristic



Start State



Goal State

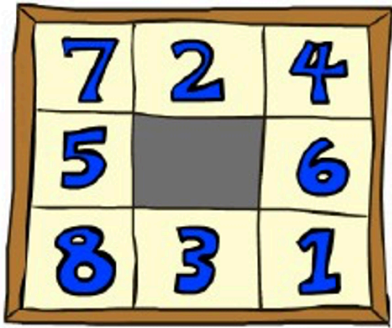


Average nodes expanded when the optimal path has...			
	...4 steps	...8 steps	...12 steps
UCS	112	6,300	3.6×10^6
A* (tiles)	13	39	227

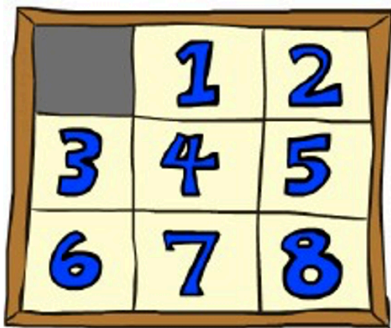
Option to grab a tile and move to original place

8 Puzzle II

- What if we had an easier 8-puzzle where any tile could slide any direction at any time, ignoring other tiles?



Start State



Goal State

- Total *Manhattan* distance

- Why is it admissible?

- $h(\text{start}) = 3 + 1 + 2 + \dots = 18$

Average nodes expanded when the optimal path has...			
	...4 steps	...8 steps	...12 steps
TILES	13	39	227
A* (Manhattan)	12	25	73

distance from current position to final position

